Supplementary Methods S1. Regional management optimization and profit evaluation

This supplementary method provides the mathematical details for the regional management optimization framework described in Section 2.4 of the main manuscript.

After model comparison, the selected yield model was used as a response function for regional management-scenario evaluation. For field i in region r, predicted yield under a candidate management vector x was expressed as:

$$\hat{Y}_{ir}(x) = f\left(X_{ir}(x)\right) \tag{S1}$$

where $f(\cdot)$ is the selected yield model and $X_{ir}(x)$ is the feature vector obtained by replacing the relevant management variables with candidate scenario values while retaining the original non-optimized background variables.

The direct decision variables were sowing day of year, plant density, N, P and K application rates, pesticide cost, irrigation electricity use and irrigation method. Irrigation electricity use was forced to zero under the no-irrigation scenario. Fertilization frequency, pesticide application frequency and irrigation frequency were inferred from related management intensities using monotonic isotonic-regression relationships fitted within each region: fertilization frequency from total nutrient input $(N+P+K)$, pesticide frequency from pesticide cost and irrigation frequency from irrigation electricity use under the corresponding irrigation mode.

For each region, candidate irrigation methods were evaluated by enumerating the irrigation modes observed in that region. For each candidate irrigation mode, the continuous decision variables were searched using differential evolution. The penalized regional objective function was:

$$J_r(x) = \mathrm{median}_{i\in r}\left\{\hat{Y}_{ir}(x)\right\} - \lambda D_r(x) \tag{S2}$$

where $J_r(x)$ is the penalized regional objective, $\mathrm{median}_{i\in r}\{\hat{Y}_{ir}(x)\}$ is the median predicted yield across fields in region r, $D_r(x)$ is the standardized nearest-neighbour distance between candidate management vector x and the observed management records in the same region, and λ is the distance-penalty weight.

The regional optimal management vector was:

$$x_r^* = \arg\max_{x\in\Omega_r} J_r(x) \tag{S3}$$

where Ω_r denotes the region-specific feasible management space. Search bounds for continuous variables were restricted to the 5th and 95th percentiles of the historical observed distribution within each region. Decision values were quantized before model evaluation: sowing date was rounded to 1 d, plant density to 500 plants ha^{-1}, N, P and K inputs to 5 kg ha^{-1}, pesticide cost to 50 CNY ha^{-1}, and irrigation electricity use to 50 kWh ha^{-1}. The implemented differential-evolution search used a random seed of 42, 120 maximum iterations, a population-size multiplier of 12 and a distance-penalty weight of 150.0.

The optimized regional yield was reported as the unpenalized median predicted yield under the selected management vector:

$$Y_r^* = \mathrm{median}_{i\in r}\left\{\hat{Y}_{ir}(x_r^*)\right\} \tag{S4}$$

The current regional baseline was defined directly from observed data, not from model predictions. Specifically, baseline yield, input cost, profit and management values were calculated as regional medians of the observed field records.

Scenario input cost was estimated using a yield-first, profit-second framework. The cost data were linked to the modelling data by field identifier. Three linear cost submodels were fitted from the observed data: sowing cost as a function of plant density, fertilization cost as a function of N, P and K application rates, and irrigation running cost as a function of irrigation electricity use. Irrigation device cost was assigned as the median observed device cost for the candidate irrigation mode. For each field, the remaining observed cost not explained by these scenario-dependent components and pesticide cost was treated as a field-level fixed residual. Thus, total input cost for field i under scenario x was calculated as:

$$C_{ir}(x) = C_i^{fixed} + C^{sow}(Density) + C^{fert}(N, P, K) + C^{pest}(PestCost) + C^{irr}(IrrElec) + C^{device}(Mode) \quad \text{(S5)}$$

where C_i^{fixed} is the fixed residual cost for field i, and the remaining terms are scenario-dependent sowing, fertilization, pesticide, irrigation running and irrigation device costs. Scenario profit was then calculated as:

$$\Pi_{ir}(x) = p \times \hat{Y}_{ir}(x) - C_{ir}(x) \quad \text{(S6)}$$

where $\Pi_{ir}(x)$ is predicted profit for field i in region r under scenario x. Regional scenario profit was summarized as the median of $\Pi_{ir}(x)$ across fields within each region.

Yield and profit gains were calculated by comparing the optimized scenario with the observed regional baseline:

$$\Delta Y_r = Y_r^* - Y_r^{base} \quad \text{(S7)}$$

$$\Delta Y_r(\%) = \frac{Y_r^* - Y_r^{base}}{Y_r^{base}} \times 100 \quad \text{(S8)}$$

$$\Delta \Pi_r = \Pi_r^* - \Pi_r^{base} \quad \text{(S9)}$$

$$\Delta \Pi_r(\%) = \frac{\Pi_r^* - \Pi_r^{base}}{\Pi_r^{base}} \times 100 \quad \text{(S10)}$$

where Y_r^{base} and Π_r^{base} are the observed baseline median yield and profit in region r, and Y_r^* and Π_r^* are the corresponding values under the optimized maximum-yield scenario.